3.1 Newton's first law of motion

Before Newton it was generally thought that a force was required to keep an object moving (see also Section 3.4.3). Newton instead postulated (on compelling experimental evidence) that an object would **continue** moving at constant speed in a straight line **unless** it was acted on by an unbalanced force.

Newton's first law of motion.
A body will continue in a state of rest, or of constant speed along a straight line, unless compelled by an **unbalanced** force to change that state.

Force is a **vector** quantity, and by the **unbalanced** or **net force** we mean the resultant force, or vector sum of all the forces acting on an object. Mathematically we write the net force

$$\boxed{\mathbf{F} = \sum \mathbf{F}_i}, \tag{21}$$

where $\sum$ is the mathematical symbol used to denote a sum of items indexed by the subscript i.

Newton's first law effectively sets the scene, because it defines the frame (or frames) of reference (i.e. coordinate system and clocks) with respect to which his remaining two laws of motion have meaning.

The class of reference frames with respect to which Newton's first law is valid are called **inertial reference frames**.

When a net force acts on an object, the object's velocity changes. The amount of change depends on the force as well as the mass of the object. If the same force acts on two objects of different mass, the more massive object will experience a smaller change in velocity. The tendency of an object with mass to resist a change in its state of motion is called the **inertia** of the object. The inertia of an object is measured by it's mass.

Inertia
Inertia is that property of a body by virtue of which it tends to persist in a state of rest or uniform motion in a straight line.

3.2 Newton's second law of motion

Newton's first law refers to a situation where there is no force. If a net force acts on an object, the object will change its state of motion according to Newton's second law.

Newton's second law of motion
If a net force acts on a body, the body will be accelerated; the magnitude of the acceleration is directly proportional to the magnitude of the net force and inversely proportional to the mass of the body, whilst the direction of the acceleration is in the direction of the net force.

In mathematical terms Newton's second law may be written as:

$$\mathbf{a} \propto \frac{\mathbf{F}}{m}$$

or

$$\mathbf{F} \propto m\mathbf{a},$$

where $\mathbf{a}$ is the magnitude of the acceleration of the mass m produced by the net force $\mathbf{F}$.

In SI units, force is defined so as to make the constant of proportionality equal to 1.

Unit of force
In the SI, the unit of force is the **newton** (N). The newton is defined as the net force which will give a mass of 1 kilogram an acceleration of $1\,\mathrm{m\,s^{-2}}$ in the direction of the force.

Using the above definition, Newton's second law takes the familiar form

$$\boxed{\mathbf{F}_{\text{net}} = m\mathbf{a}} \tag{22}$$

where $\mathbf{F}$ is the **net force** in newtons (N), m in kilograms and $\mathbf{a}$ in $\mathrm{m\,s^{-2}}$.

3.2.1 Free-body diagrams

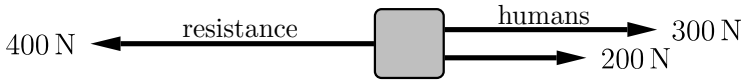
A **free-body diagram** is a diagram that represents an object and the forces acting on the object. The forces are represented by arrows (since force is a vector quantity) with length proportional to the magnitude of the force and the direction of the arrow indicating the direction of the force. A free-body diagram should **always** be drawn when a problem involves Newton's second law.

Example 14: Free-body diagram

Suppose two people push a car along a horizontal road. One person applies a force of 300 N and the other a force of 200 N. The car has a mass of 1200 kg and the total force due to resistance is 400 N and acts in the opposite direction to the forces exerted by the people pushing. Draw a free-body diagram that shows the horizontal forces on the car and find the acceleration of the car.

Solution:

We need to use Newton's second law. However we first need to find the resultant force. The free-body diagram below shows the car as a square and the arrows representing the forces.



Since the forces all act along one direction (the x direction say), the resultant force will also be in this direction, as will the acceleration. We choose the positive direction to the right, thus the net force is

$$F = \sum F_i = +300\,\mathrm{N} + 200\,\mathrm{N} - 400\,\mathrm{N} = +100\,\mathrm{N}.$$

The positive sign indicates that the force, **and hence the acceleration**, is in the direction we chose to be to the right.

The acceleration may now be determined from Equation (22):

$$a = \frac{F}{m} = \frac{100\,\mathrm{N}}{1200\,\mathrm{kg}} = 0.083\,\mathrm{m\,s^{-2}}.$$

3.3 Newton's third law

Newton's third law
If body A exerts a force on body B, body B exerts a force equal in magnitude and opposite in direction on body A.

Mathematically:

$$\mathbf{F}_{\mathrm{AB}} = -\mathbf{F}_{\mathrm{BA}} \tag{23}$$

If two people pull on spring scales hooked together, no matter how hard each person tries to pull, the readings on the two scales will be the same. Likewise, if a ball is hit with a bat, there is not only a force exerted by the bat on the ball, but also a force, which is the same in magnitude but opposite in direction, exerted by the ball on the bat. To walk, a person exerts a force on the earth. Consistent with Newton's third law, the earth exerts an oppositely directed force of equal magnitude on the person's foot and causes the person to move forward. Note that in Newton's third law and in all the above examples: **the two forces act on different bodies**.

3.4 Types of forces

There are four known types of forces in Nature.

1. **Gravitational force**
   Gravitation is the weakest of the known forces. It is also the only force that is purely attractive. Weight is a gravitational force (see Section 3.4.1).

2. **Electromagnetic force**
   Most forces we encounter in daily life are electromagnetic forces, they arise from the interaction of the electrically charged particles that make up atoms and molecules. Important examples are:

   - Normal contact forces like collisions and throwing of objects.
   - Tension forces such as surface tension and the tension in stretched strings.
   - Compressive forces such as those in springs or in a hydraulic press.
   - Frictional forces such as air drag on a skydiver and the grip on shoes (see Section 3.4.3).

   Many, if not most forces we encounter are combinations of forces.

3. **Weak nuclear force**
   A manifestation of the electromagnetic force that plays a role in the radioactive decay of atoms.

4. **Strong nuclear force**
   Plays a role in the interactions in nuclei of atoms.

3.4.1 The gravitational force and weight

Every particle in the universe attracts every other particle. The magnitude of the force acting on each of two particles of mass m_1 and m_2, separated by a distance r, is given by

$$\boxed{F = \frac{Gm_1m_2}{r^2}}, \tag{24}$$

where $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$ is the **universal gravitational constant**.

By a particle we understand something so small that it may be regarded as a mathematical point. Although Equation (24) is for 'point' particles, it can be used with good accuracy when the masses are small compared to the distance separating them. For objects that are not particles, r in Equation (24) is the distance between the **centres** of the objects.

The **weight** of an object is the gravitational force the earth exerts on it.

The weight always acts downward, towards the centre of the earth. An object will not necessarily weigh the same on another planet.

If the mass of an object is m and the acceleration due to gravity is g, then its weight on earth is given by

$$W = mg = \frac{GM_\mathrm{E}m}{R_\mathrm{E}^2},$$

where M_E and R_E are the mass and radius of the earth respectively. Here we used Newton's second law (Equation (22)) with the acceleration $a = g$. Since the mass m of the object appears on both sides of the equation, the acceleration due to gravity on a planet of mass M and radius R is given by

$$g = \frac{GM}{R^2}. \tag{25}$$

Example 15: The mass of the earth

Calculate the mass of the earth given that the radius of the earth is $R_\mathrm{E} = 6.38 \times 10^6\,\mathrm{m}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$.

Solution:

We rearrange Equation (25) with M the subject. Then

$$\begin{aligned} M_\mathrm{E} = \frac{gR_\mathrm{E}^2}{G} &= \frac{9.8\,\mathrm{m\,s^{-2}} \times \left(6.38 \times 10^6\,\mathrm{m}\right)^2}{6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}} \\ &= 5.98 \times 10^{24}\,\mathrm{kg}. \end{aligned}$$

For distances greater than the earth's radius (as in the case of a satellite), we must add the height above the earth's surface to the radius.

Example 16: Acceleration due to gravity for a satellite in orbit

Determine the acceleration due to gravity for a satellite in orbit 200 km above the surface of the earth. Use the same data given in Example 15.

Solution:

We again use Equation (25) with the following data: $M = M_{\mathrm{E}} = 5.98 \times 10^{24}\,\mathrm{kg}$
$R = R_{\mathrm{E}} + h = (6.38 \times 10^{6} + 200 \times 10^{3})\,\mathrm{m}$ and
$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$. Hence

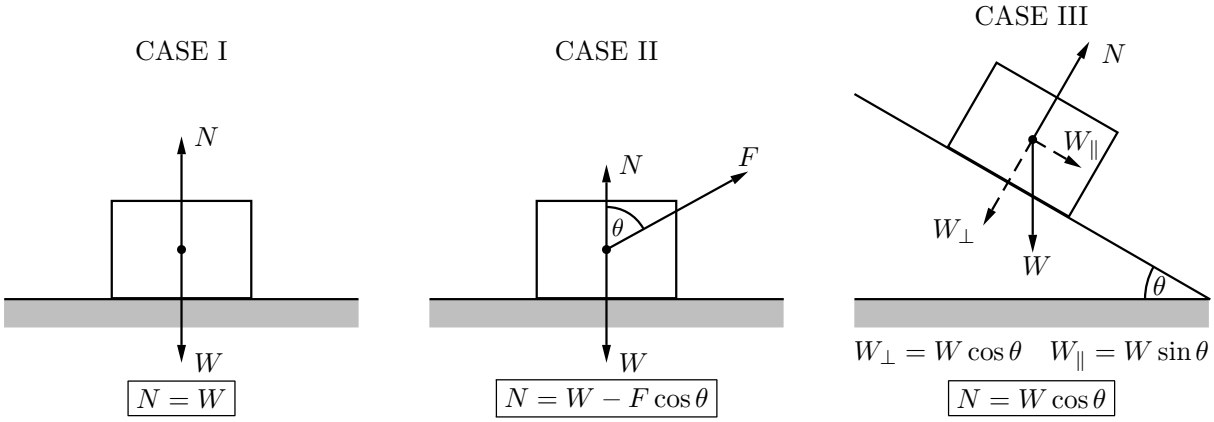
$$g = \frac{GM}{R^2} = \frac{6.67 \times 10^{-11} \times 5.98 \times 10^{24}}{(6.38 \times 10^{6} + 200 \times 10^{3})^2} = 9.2\,\mathrm{m\,s^{-2}}.$$

3.4.2 The Normal force

Normal in a mathematical sense means **perpendicular**. An object resting on a table for instance exerts a force equal to it's weight on the surface of the table. By Newton's third law, the table exerts an equal and opposite force on the object.

The **normal force** $\mathbf{F}_{\mathrm{N}}$ is the force, or component of a force, that a surface exerts on an object in contact with it.

The normal force N is often equal to the weight W of a body — but it is **not necessarily so**. Consider the following cases:



Example 17: Weight and the normal force in an elevator

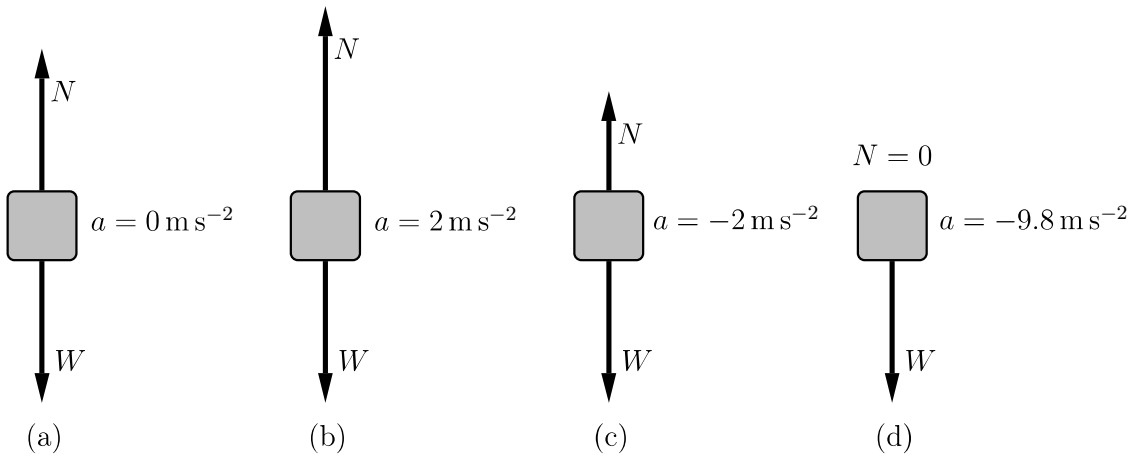
Find the apparent weight of a person whose mass is 60 kg in an elevator, when the elevator

(a) is stationary,

(b) accelerating upward at $2\,\mathrm{m\,s^{-2}}$,

(c) accelerating downward at $2\,\mathrm{m\,s^{-2}}$, and

(d) in free-fall.



Solution:

Imagine a person standing on a scale in the elevator. The apparent weight of the person is the normal force exerted by the scale on the man (the reading on the scale). Hence we must find the net force in each case and use Newton's second law with the acceleration a given by the acceleration of the lift. In the diagrams above, W is the weight of the person (unchanged in each case) and N the normal force. The net force on the person is the vector sum of the normal force and the weight. If we regard the upward direction as positive, then

$$F = -W + N,$$

where

$$W = mg = 60 \times 9.8 = 588\,\text{N}.$$

Using Newton's second law ($F = ma$), the normal force is

$$N = ma + W.$$

(a) The acceleration is zero, hence

$$N = ma + W = 0 + 588 = 588\,\text{N}.$$

(b) Here $a = 2\,\text{m}\,\text{s}^{-2}$, therefore

$$N = ma + W = 60 \times 2 + 588 = 120 + 588 = 708\,\text{N}.$$

(c) The acceleration is now $a = -2\,\text{m}\,\text{s}^{-2}$, hence

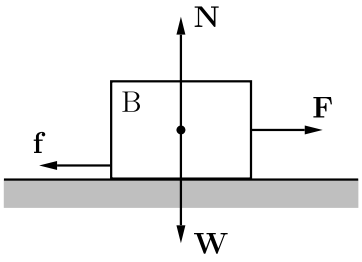
$$N = ma + W = 60 \times (-2) + 588 = -120 + 588 = 468\,\text{N}.$$

(d) In free-fall, the lift (and person inside) is accelerating downwards at $9.8\,\text{m}\,\text{s}^{-2}$. Thus $a = -9.8\,\text{m}\,\text{s}^{-2}$ and

$$N = ma + W = 60 \times (-9.8) + 588 = -588 + 588 = 0\,\text{N}.$$

3.4.3 Friction

The force that opposes the motion of one surface moving over another, with which it is in contact, is called the **force of friction**. Its magnitude depends on the materials of which the two surfaces are made, as well as on the force pressing them together. Some of the energy put into machines is transformed into heat energy because of the friction between moving parts. The heat may cause serious damage in addition to being wasteful. We cannot get rid of friction entirely but we can reduce it considerably by suitable choice of surfaces and by using lubrication.



Consider a force **F** applied to a block B on a horizontal surface S. If F is slowly increased from zero, the body remains at rest until F reaches a certain value, after which B accelerates in the direction of **F** (to the right in the diagram alongside).

As the force F increases, the **force of static friction** f_s increases and 'adjusts itself' to always exactly cancel F. The **maximum value** of the force of static friction is $f_\mathrm{s}\,(\mathrm{max})$. Experimentally we find that

$$\boxed{f_\mathrm{s}\,(\mathrm{max}) = \mu_\mathrm{s} N}, \tag{26}$$

where N is the normal force between the two surfaces in contact and μ_s, the **coefficient of static friction**, is a physical constant for the pair of surfaces in contact.

When the applied force F is larger than $\mu_\mathrm{s}N$, there is a resultant force in the direction of **F** and the body will start to slip and accelerate in the direction of **F**. Once the block starts to slide, the frictional force drops below μ_s. We now talk about the **force of kinetic (or sliding) friction** f_k. Experimentally it is found that

$$\boxed{f_\mathrm{k} = \mu_\mathrm{k} N}, \tag{27}$$

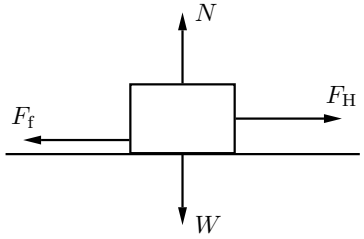
where μ_k, the coefficient of kinetic friction, is usually slightly less than μ_s. It is an experimental fact that μ_s depends only on the nature of the sliding surfaces; it is independent of the area of contact or the relative speed of the surfaces.

Example 18: Block on a horizontal surface with friction

A block which has a mass of 100 kg rests on a rough horizontal floor. The coefficient of sliding friction between the block and the floor is 0.25. Calculate the horizontal force F_H which would be required to move the block along the floor with constant velocity.

Solution:

When the block moves with constant velocity $a = 0$. Therefore, since $F = ma$, and $a = 0$, the total unbalanced force F must be zero. Taking the positive direction to the right,



$$F_\mathrm{H} - F_\mathrm{f} = 0.$$

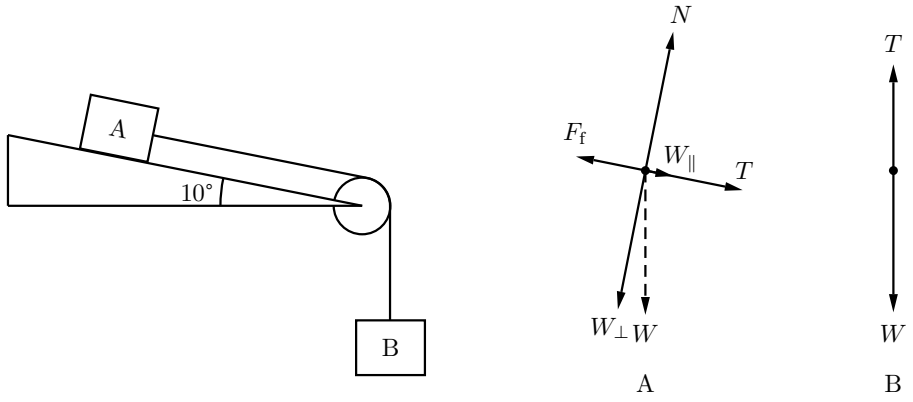
Also, $F_\mathrm{f} = \mu N = mg\mu$, therefore

$$F_\mathrm{H} = mg\mu = 100 \times 9.8 \times 0.25 = 245\,\mathrm{N}.$$

Example 19: Block on an inclined plane with friction

A body of mass 20 kg rests on a plane surface AB inclined at 10° to the horizontal, B being lower than A. The mass is connected by a light string which passes over a pulley at B, to another mass of 20 kg that hangs freely below B. If the coefficient of sliding friction between the body and the surface of the plane is 0.40, calculate the acceleration with which these bodies would move, and the tension in the string connecting them.

Solution:



Consider the acceleration of both masses and choose the direction of motion as positive. The net force in the direction of motion is then

$$W - T + T + W_\parallel - F_f,$$

where the frictional force $F_f = \mu N = \mu mg\cos 10°$. The total mass $m = 2m = 40\,\text{kg}$. Newton's second law then gives

$$mg + mg\sin 10° - 0.4mg\cos 10° = 2ma$$

which gives $a = 3.82\,\text{m}\,\text{s}^{-2}$. (Note that because the masses are equal, we do not need to know the mass for this calculation.)

The tension T can be obtained by considering the motion of the mass B. Newton's second law gives $W - T = ma$, hence

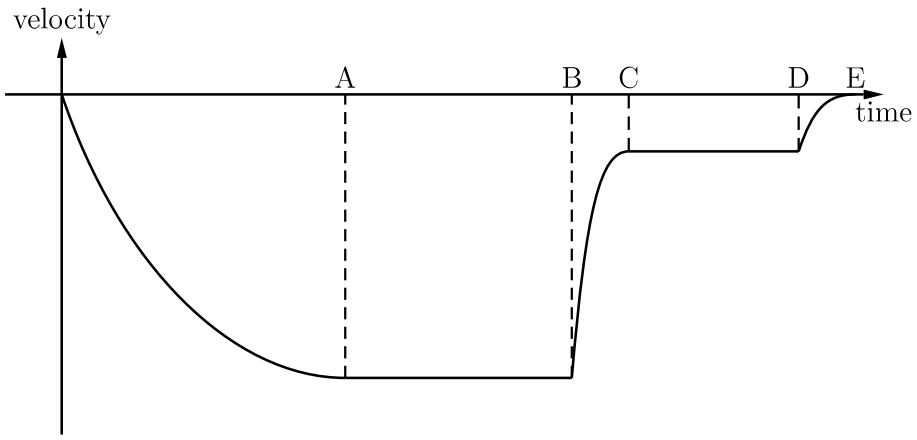
$$T = W - ma = 20 \times 9.8 - 20 \times 3.82 = 119.6\,\text{N}.$$

A considerably smaller force called **rolling** friction is sufficient to keep one body moving against another if there are hard rollers or balls between two surfaces. It is important that the metal surfaces of roller or ball bearings that come into contact should be really hard. If one of the surfaces is not hard then the rolling friction might well be more than the sliding friction; it is for this reason that aircraft landing on soft snow fit skis in place of wheels.

An important example of Newton's first law is the case of an object falling through a medium. When it first starts to fall, it speeds up because its weight is bigger than the upthrust on it. But the dragging force on it increases as its velocity increases, and a stage can be reached when the upthrust plus dragging force (upwards) is as large as the weight (downwards). There is then no unbalanced force and the object continues to fall with the velocity it had reached — a constant velocity known as the **terminal velocity**.

Example 20: Velocity time graph for a skydiver

A skydiver jumps from an aeroplane. Sketch a velocity–time graph for the vertical motion of the skydiver indicating where the skydiver reaches terminal velocity, opens her parachute and lands.



Solution:

The origin represents the moment the skydiver jumps from the aeroplane. Her initial velocity in the vertical direction is zero at this point. Her acceleration on the other hand is equal to the acceleration due to gravity. Since we take the upward direction as positive, the acceleration is negative. (The slope of the graph gives the acceleration. If we draw a tangent to the curve at the origin, the slope of this line is negative). As the downward velocity increases, the **magnitude** of the acceleration decreases as the frictional drag due to the air increases. The force due to friction is in the **opposite** direction to the velocity. At A the skydiver reaches terminal velocity — here the force due to the air resistance is equal and opposite to the force due to gravity. The slope of the graph between A and B is zero and the velocity remains the same. At B she opens her parachute. Her acceleration here is positive and is a maximum as she opens her parachute, decreasing to zero as she again reaches terminal velocity at C. The segment between D and E is where she reaches the ground.

3.4.4 Tension

If two people pull on either end of a rope there will be a certain tension in the rope. The force experienced by each person will be the same and will equal the tension in the rope. Figure 8 depicts the free-body diagram for this scenario. Each end of the rope provides the reaction



Figure 8: Tension in a rope

force on the person pulling at that end, as required by Newton's third law. The force of the people pulling on the rope in effect gets transmitted through the rope.

3.5 Application of Newton's laws

In this section we discuss applications of Newton's laws to various systems. In Section 3.5.2 we consider systems that are in equilibrium and in Section 3.5.3 some examples of non-equilibrium situations are discussed.

3.5.1 Guidelines for solving problems involving Newton's laws

The following points must be remembered when the relationship $F = ma$ is used:

1. The mass in Newton's second law ($F = ma$) represents the **total mass** accelerated by the **net force**.
2. F represents the **net force in the direction of motion**. If the accelerated mass is acted upon by a number of forces, the total net component of the forces in the direction of the motion must be calculated.

The following systematic approach to problems in which the relationship $F = ma$ has to be used may be useful:

1. Draw a diagram representing the **general** situation.
2. Select **one object** from the situation whose motion is to be analysed and draw a **free-body diagram** for this object. For this, the object is removed from its environment, together with **all the forces** exerted on it by bodies with which it interacts.
3. Select a convenient origin and orientation of the coordinate axes.
4. Write an expression for the **net force**.
5. Apply Newton's second law.

3.5.2 Equilibrium applications

Equilibrium
An object is in equilibrium when it has zero acceleration.

When the acceleration of an object is zero, the net force on the object is zero by Newton's first law. Thus when an object is in equilibrium in two dimensions, we must have

$$\sum F_{\mathrm{x}} = 0 \tag{28a}$$

$$\sum F_{\mathrm{y}} = 0. \tag{28b}$$

Examples of systems in equilibrium include a book lying on a table, a lamp hanging from a cord or a vehicle moving at constant velocity.

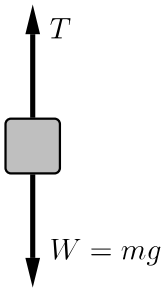
Example 21: Tension in a cord, one dimension (equilibrium case)

A lamp is suspended from the ceiling by a cord. If the lamp has a mass of 5 kg, determine the tension in the cord.

Solution:

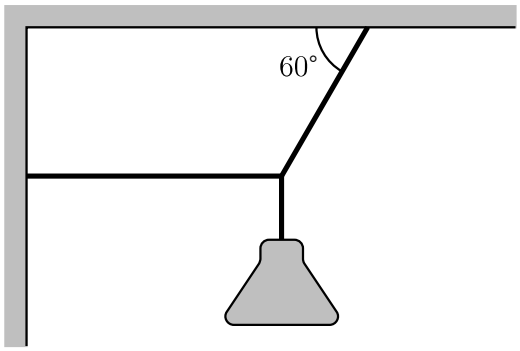
Since the system is in equilibrium, we use Equation (28b) to find the net force. $T - W = 0$ gives

$$T = W = mg = 5\,\mathrm{kg} \times 9.8\,\mathrm{m\,s^{-2}} = 49\,\mathrm{N}.$$



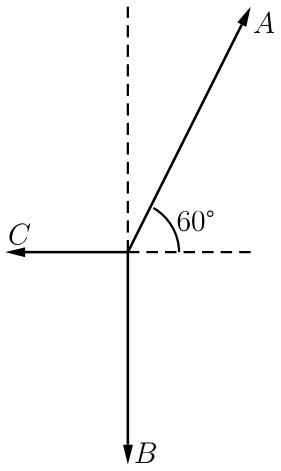
Example 22: Tension in a cord, two dimensions (equilibrium case)

A lamp is suspended by three cords as depicted in the diagram below. The cord attached to the ceiling makes an angle of 60° with the ceiling and the cord attached to the wall is stretched horizontally. If the lamp has a mass of 5 kg, determine the tensions in the cords.



Solution:

Since the forces (tensions in the cords) do not act in the same direction, we will need to resolve the components of the forces in the x and y directions. First we construct a free-body diagram representing the forces acting at the intersection of the cords (the magnitudes of the forces are labelled A, B and C).



The tension B is simply equal to the weight of the lamp. Hence

$$B = W = mg = 5 \times 9.8 = 49\,\text{N}.$$

Equating the x and y components of A, B and C according to Equations (21), we have

$$A_x + B_x + C_x = 0 \quad \text{and} \quad A_y + B_y + C_y = 0,$$

where

$$\begin{aligned} A_x &= A\cos 60^\circ \quad \text{and} \quad A_y = A\sin 60^\circ, \\ B_x &= 0 \quad \text{and} \quad B_y = -B, \\ C_x &= -C \quad \text{and} \quad C_y = 0. \end{aligned}$$

Thus, equating first the y components:

$$A\sin 60^\circ - 49 + 0 = 0, \quad \text{which gives} \quad A = 56.6\,\text{N}.$$

Equating the x components:

$$A\cos 60^\circ + 0 - C = 0, \quad \text{which gives} \quad C = 28.3\,\text{N}.$$

3.5.3 Non-equilibrium applications

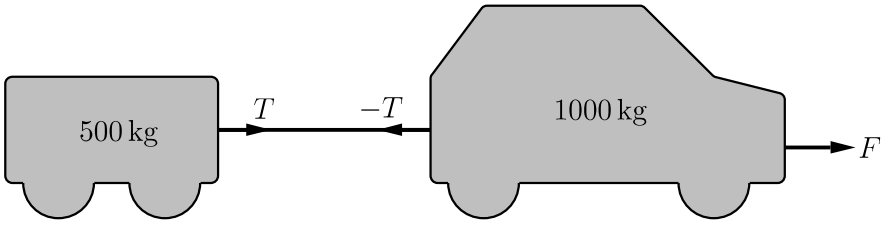
When an object is not in equilibrium, there are unbalanced forces acting on the object and hence the net force is non-zero. The approach to solving non-equilibrium problems is almost identical to the approach used to treat equilibrium problems. Instead of equating the net force to zero as in Equations (28), we must use Newton's second law. Thus for an accelerating object in two dimensions

$$\sum F_{\mathrm{x}} = ma_x \tag{29a}$$

$$\sum F_{\mathrm{y}} = ma_y. \tag{29b}$$

Example 23: The tension in a rope (non-equilibrium)

Suppose the magnitude of the net force accelerating a car and trailer is $F = 3000\,\mathrm{N}$. The mass of the car is 1000 kg and the mass of the trailer is 500 kg. Determine the acceleration of the car and trailer, and the tension in the rope. Assume the mass of the rope is negligible.



Solution:

The acceleration may be determined by applying Newton's second law to the whole system (the car and trailer). We are given the net force and we know the total mass of the system. Thus

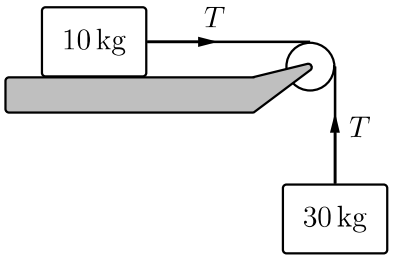
$$a = \frac{F}{m} = \frac{3000}{1000 + 500} = 2\,\mathrm{m\,s^{-2}}.$$

Newton's second law may also be applied to the trailer by itself. Here the net force in the horizontal direction is the tension T and the mass of the system is the mass of the trailer. The acceleration was found above. Hence

$$F = T = ma = 500 \times 2 = 1000\,\mathrm{N}.$$

Example 24: Objects connected by a rope

A block of mass 10 kg on a table is attached to a block of mass 30 kg by a rope passing over a pulley as shown in the diagram alongside. Ignoring all frictional effects and assuming the pulley to be massless, find (a) the acceleration of the two blocks and (b) the tension in the cord. (Take $g = 10\,\mathrm{m\,s^{-2}}$.)



Solution:

(a) The net force available to accelerate the system is due to the 30 kg mass (the forces on the 10 kg block in the y direction are equal and opposite). Hence

$$F = W = -mg = -30\,\text{kg} \times 10\,\text{m}\,\text{s}^{-2} = -300\,\text{N}.$$

The total mass of the system is the mass of the two blocks, $m_{\text{TOT}} = 10\,\text{kg} + 30\,\text{kg} = 40\,\text{kg}$. We can now use Newton's second law to find the acceleration.

$$a = \frac{F}{m} = \frac{-300\,\text{N}}{40\,\text{kg}} = -7.5\,\text{m}\,\text{s}^{-2}.$$

(b) the only unbalanced force on the 10 kg mass is due to the tension T in the rope. Using Newton's second law with $a = 7.5\,\text{m}\,\text{s}^{-2}$, since the 10 kg block is accelerated in the positive x direction,

$$T = ma = 10\,\text{kg} \times 7.5\,\text{m}\,\text{s}^{-2} = 75\,\text{N}.$$

3.5.4 Motion on a smooth inclined plane

When a block of mass m is placed on a smooth frictionless inclined plane, as shown in Figure 9, the block will be accelerated down the plane. The force in the direction of motion which gives rise to the acceleration of the mass is the component of the weight of the body down the plane. The weight W of the body (which is a force acting vertically downwards) can be resolved into components acting along the plane and perpendicular to the plane (see also Section 1.3.5).

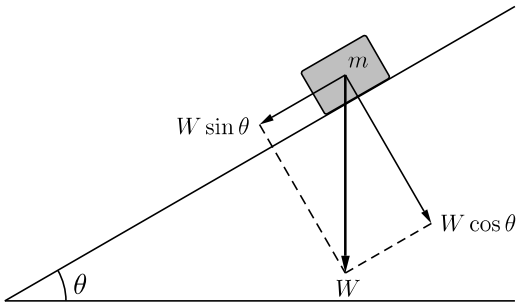


Figure 9: An object on a smooth inclined plane.

If the plane is inclined at an angle θ to the horizontal, then the component of the weight parallel to the plane is

$$W_{\parallel} = W \sin\theta,$$

and the component perpendicular to the plane is

$$W_{\perp} = W \cos\theta.$$

Example 25: Motion on an inclined plane

Show that the acceleration of a sliding body down a frictionless plane is **independent of the mass of the body**.

Solution:

Consider a body with mass m which is placed on a frictionless plate inclined at an angle θ to the horizontal as in Figure 9. The net force in the direction of the motion is

$$F_{\parallel} = W \sin\theta.$$

The total mass accelerated is m. Using Newton's second law, we have

$$F = ma = F_{\parallel} = W \sin\theta = mg \sin\theta,$$

and hence

$$a = g \sin\theta,$$

which is independent of the mass m.

3.6 The centripetal force

For an object moving at constant speed v in a circle of radius r the centripetal acceleration is given by

$$a = \frac{v^2}{r}$$

(see Section 2.5.2). By Newton's first law, if an object is accelerating, there must be a force acting on it. For circular motion this force is known as the **centripetal force**. The centripetal force may take different forms. For instance, for a car travelling in a circular path on a horizontal surface the centripetal force is the frictional force between the tyres and the surface; for a satellite in orbit around a planet the centripetal force is the gravitational force; for an object at the end of a string spun in a circular path, the tension in the string provides the centripetal force. Newton's second law takes the form

$$\sum F_{\text{net}} = m\frac{v^2}{r}, \tag{30}$$

where the resultant force causing the circular motion $\sum F_{\text{net}}$ is called the centripetal force. The centripetal force is directed towards the centre of rotation (in the same direction as the acceleration).

3.7 Satellites in circular orbits

Consider an object of mass m (e.g. a satellite) moving in a circular orbit (radius r) at a constant speed v_0 around the earth. Newton's law of gravitation, the second law $F = ma$ with $a = v_0^2/r$ (see Equations (24), (22) and (20)), then gives

$$\frac{GmM}{r^2} = m\left(\frac{v_0^2}{r}\right),$$

or

$$v_0^2 = \frac{GM}{r}. \tag{31}$$